

入院 RAR 指数与胰腺癌患者 28 天内全因死亡：基于 MIMIC-IV 与 eICU 的双数据库预测与预后研究

一、研究摘要

本研究在两个相互独立的重症监护数据库中评估红细胞分布宽度与白蛋白的比值（RAR）对胰腺癌患者 28 天内全因死亡的预后价值，并构建可解释的死亡风险预测模型。MIMIC-IV 队列（308 例，死亡 101 例，32.8%）作为训练集，eICU 队列（141 例，死亡 19 例，13.5%）作为外部验证集。

主要发现可归纳为四点。第一，单因素分析中 RAR 在训练集（OR 1.18, 95% CI 1.06–1.32, $p = 0.003$ ）与验证集（OR 1.70, 95% CI 1.29–2.23, $p < 0.001$ ）均与死亡显著相关，且在验证集中效应更强。第二，限制性立方样条（RCS）显示 RAR 与死亡呈近似线性的剂量–反应关系，非线性检验均不显著（ P for nonlinearity > 0.4 ），经多模型调整后关联依然稳健。第三，基于 10 个临床特征训练的 15 种机器学习模型在验证集中以 TablCL v2（AUC 0.852）、XGBoost（0.842）、TabPFN（0.836）、RSVM（0.834）判别力较优。第四，SHAP 分析显示 RAR 是模型最重要的预测变量（平均 | SHAP | 0.642），高于尿素氮、钾、氯等指标。

结论：入院 RAR 是胰腺癌患者短期院内死亡的独立、近似线性且可解释的预后标志物；以 RAR 为核心的多特征机器学习模型具备可接受的外部判别力。本报告为阶段性结果，最终结论有待敏感性分析与外部数据的进一步验证。

二、数据

3.1 数据来源与队列

研究对象来自 MIMIC-IV 与 eICU 两个公开重症监护数据库。MIMIC-IV 队列作为模型训练与变量筛选的训练集，eICU 队列作为外部验证（测试）集。按既定纳入排除标准获得胰腺癌患者，剔除关键变量缺失过多者后，训练集 308 例、验证集 141 例，合计 449 例。考虑到数据库变量存在不同程度的缺失，分析前先对两个数据库分别进行多重插补。

3.2 暴露变量、结局与协变量

- 核心暴露：RAR，定义为 RDW 与白蛋白的比值。

- **结局：**28 天院内全因死亡。

- **候选协变量：**人口学（年龄、性别、种族）、生命体征（心率 HR、收缩压 NBPS、舒张压 NBPD、呼吸频率 RR、体温）、实验室指标（白细胞 WBC、血小板、血红蛋白、钠、钾、钙、氯、总胆红素、ALT、AST、尿素氮、肌酐）、合并症（高血压、2 型糖尿病、1 型糖尿病、心力衰竭、心肌梗死、慢性肾病、肺炎、卒中、高脂血症、慢阻肺）。

三、结果（含图表解读）

3.1 多重插补：Table S1

Table S1—MIMIC. Baseline characteristics of patients before and after multiple imputation1		BeforeMI (N=308)	AfterMI (N=308)	
Age	Mean (SD)	70 (± 11)	70 (± 11)	1
Gender				
	Female	133 (43 %)	133 (43 %)	1
	Male	175 (57 %)	175 (57 %)	
Race				
	Asian	16 (5 %)	16 (5 %)	1
	Black	29 (9 %)	29 (9 %)	
	Hispanic	8 (3 %)	8 (3 %)	
	Other	29 (9 %)	29 (9 %)	
	White	226 (73 %)	226 (73 %)	
Weight	Mean (SD)	75 (± 20)	75 (± 20)	1
	Missing	2 (0.6%)	0 (0%)	
HR	Mean (SD)	96 (± 20)	96 (± 20)	1
NBPS				
	Mean (SD)	110 (± 22)	110 (± 22)	1
	Missing	5 (1.6%)	0 (0%)	
NBPd				
	Mean (SD)	65 (± 16)	65 (± 16)	0.997
	Missing	5 (1.6%)	0 (0%)	
RR				
	Mean (SD)	20 (± 5.7)	20 (± 5.7)	0.989
	Missing	5 (1.6%)	0 (0%)	
Temperature				
	Mean (SD)	37 (± 0.58)	37 (± 0.60)	0.983
	Missing	6 (1.9%)	0 (0%)	
WBC	Mean (SD)	14 (± 9.5)	14 (± 9.5)	1
PlateletCount	Mean (SD)	200 (± 120)	200 (± 120)	1
Hemoglobin				
	Mean (SD)	9.7 (± 1.9)	9.7 (± 1.9)	0.999
	Missing	2 (0.6%)	0 (0%)	
Sodium				
	Mean (SD)	140 (± 5.2)	140 (± 5.3)	0.998
	Missing	1 (0.3%)	0 (0%)	
Potassium	Mean (SD)	4.1 (± 0.77)	4.1 (± 0.77)	1
CalciumTotal	Mean (SD)	8.0 (± 0.86)	8.0 (± 0.86)	1
Chloride	Mean (SD)	100 (± 6.6)	100 (± 6.6)	1
BilirubinTotal				
	Mean (SD)	3.2 (± 4.8)	3.2 (± 4.9)	0.999
	Missing	9 (2.9%)	0 (0%)	
ALT				
	Mean (SD)	110 (± 250)	110 (± 250)	0.974
	Missing	8 (2.6%)	0 (0%)	
AST				
	Mean (SD)	140 (± 280)	140 (± 280)	0.971
	Missing	7 (2.3%)	0 (0%)	
UreaNitrogen				
	Mean (SD)	29 (± 23)	29 (± 24)	0.998
	Missing	1 (0.3%)	0 (0%)	
Creatinine	Mean (SD)	1.4 (± 1.4)	1.4 (± 1.4)	1
Hypertension				
	No	162 (53 %)	162 (53 %)	1
	Yes	146 (47 %)	146 (47 %)	
T2DM				
	No	158 (51 %)	158 (51 %)	1
	Yes	150 (49 %)	150 (49 %)	
T1DM				
	No	307 (100 %)	307 (100 %)	1
	Yes	1 (0 %)	1 (0 %)	
Heart Failure				
	No	261 (85 %)	261 (85 %)	1
	Yes	47 (15 %)	47 (15 %)	
Myocardial Infarction				
	No	289 (94 %)	289 (94 %)	1
	Yes	19 (6 %)	19 (6 %)	
CKD				
	No	256 (83 %)	256 (83 %)	1
	Yes	52 (17 %)	52 (17 %)	
Pneumonia				
	No	231 (75 %)	231 (75 %)	1
	Yes	77 (25 %)	77 (25 %)	
Stroke				
	No	294 (95 %)	294 (95 %)	1
	Yes	14 (5 %)	14 (5 %)	
Hyperlipidemia				
	No	196 (64 %)	196 (64 %)	1
	Yes	112 (36 %)	112 (36 %)	
COPD				
	No	293 (95 %)	293 (95 %)	1
	Yes	15 (5 %)	15 (5 %)	
DN	Mean (SD)	0.33 (± 0.47)	0.33 (± 0.47)	1
futime	Mean (SD)	23 (± 8.2)	23 (± 8.2)	1
RAR	Mean (SD)	6.7 (± 2.2)	6.7 (± 2.2)	1

Table S1—eICU. Baseline characteristics of patients before and after multiple imputation1		BeforeMI (N=141)	AfterMI (N=141)	
Age	Mean (SD)	66 (± 10)	66 (± 10)	1
Gender				
	Female	73 (52 %)	73 (52 %)	1
	Male	68 (48 %)	68 (48 %)	
Race				
	Asian	3 (2 %)	3 (2 %)	1
	Black	17 (12 %)	17 (12 %)	
	Hispanic	11 (8 %)	11 (8 %)	
	Other	3 (2 %)	3 (2 %)	
	White	107 (76 %)	107 (76 %)	
Weight	Mean (SD)	78 (± 23)	78 (± 23)	1
	Missing	1 (0.7%)	0 (0%)	
HR				
	Mean (SD)	92 (± 17)	92 (± 16)	0.987
	Missing	12 (8.5%)	0 (0%)	
NBPS				
	Mean (SD)	120 (± 25)	120 (± 25)	0.961
	Missing	12 (8.5%)	0 (0%)	
NBPd				
	Mean (SD)	64 (± 16)	65 (± 17)	0.95
	Missing	12 (8.5%)	0 (0%)	
RR				
	Mean (SD)	19 (± 5.1)	19 (± 5.0)	0.993
	Missing	12 (8.5%)	0 (0%)	
Temperature				
	Mean (SD)	37 (± 0.72)	37 (± 0.72)	0.999
	Missing	2 (1.4%)	0 (0%)	
WBC	Mean (SD)	13 (± 7.6)	13 (± 7.6)	1
PlateletCount	Mean (SD)	200 (± 110)	200 (± 110)	1
Hemoglobin	Mean (SD)	9.9 (± 1.9)	9.9 (± 1.9)	1
Sodium	Mean (SD)	140 (± 4.6)	140 (± 4.6)	1
Potassium	Mean (SD)	4.1 (± 0.67)	4.1 (± 0.67)	1
CalciumTotal	Mean (SD)	7.9 (± 0.68)	7.9 (± 0.68)	1
Chloride	Mean (SD)	110 (± 6.1)	110 (± 6.1)	1
BilirubinTotal				
	Mean (SD)	2.7 (± 3.9)	2.7 (± 3.8)	1
	Missing	3 (2.1%)	0 (0%)	
ALT				
	Mean (SD)	150 (± 380)	150 (± 370)	0.997
	Missing	3 (2.1%)	0 (0%)	
AST				
	Mean (SD)	240 (± 650)	230 (± 650)	0.993
	Missing	3 (2.1%)	0 (0%)	
UreaNitrogen				
	Mean (SD)	25 (± 22)	25 (± 23)	0.995
	Missing	1 (0.7%)	0 (0%)	
Creatinine	Mean (SD)	1.3 (± 1.3)	1.3 (± 1.3)	1
Hypertension				
	No	117 (83 %)	117 (83 %)	1
	Yes	24 (17 %)	24 (17 %)	
T2DM				
	No	137 (97 %)	137 (97 %)	1
	Yes	4 (3 %)	4 (3 %)	
T1DM				
	No	139 (99 %)	139 (99 %)	1
	Yes	2 (1 %)	2 (1 %)	
Heart Failure				
	No	135 (96 %)	135 (96 %)	1
	Yes	6 (4 %)	6 (4 %)	
Myocardial Infarction				
	No	139 (99 %)	139 (99 %)	1
	Yes	2 (1 %)	2 (1 %)	
CKD				
	No	132 (94 %)	132 (94 %)	1
	Yes	9 (6 %)	9 (6 %)	
Pneumonia				
	No	131 (93 %)	131 (93 %)	1
	Yes	10 (7 %)	10 (7 %)	
Stroke				
	No	140 (99 %)	140 (99 %)	1
	Yes	1 (1 %)	1 (1 %)	
Hyperlipidemia				
	No	134 (95 %)	134 (95 %)	1
	Yes	7 (5 %)	7 (5 %)	
COPD				
	No	133 (94 %)	133 (94 %)	1
	Yes	8 (6 %)	8 (6 %)	
DN	Mean (SD)	0.13 (± 0.34)	0.13 (± 0.34)	1
futime	Mean (SD)	10 (± 7.0)	10 (± 7.0)	1
RAR	Mean (SD)	7.0 (± 2.0)	7.0 (± 2.0)	1

Table S1. Baseline characteristics of patients before and after multiple imputation (MIMIC、eICU 分库)。

结果：两库中绝大多数变量缺失比例很低。以 MIMIC 为例，体重缺失 0.6%，生命体征（NBPS、NBPD、RR）缺失约 1.6%，其余实验室指标缺失更少。插补前后各变量的分布几乎一致，差异均无统计学意义（多数 $p > 0.99$ ）。

解读：插补未改变数据的整体结构，说明缺失模式接近随机缺失、插补结果可信。这为后续所有分析提供了完整数据矩阵，同时排除了"插补本身引入偏倚"的担忧。

3.2 基线特征：Table 1

Table 1. Baseline Characteristics (MIMIC、eICU 分库，按生存/死亡分层)。

Table 1-MIMIC.Baseline Characteristics

Variable		Overall N = 308	Non-survivor N = 101	Survivor N = 207	p-value
Age		69.68 (11.28)	70.76 (10.81)	69.15 (11.50)	0.237
Gender					0.169
	Female	133.00 (43.18%)	38.00 (37.62%)	95.00 (45.89%)	
	Male	175.00 (56.82%)	63.00 (62.38%)	112.00 (54.11%)	
Race					0.787
	Asian	16.00 (5.19%)	4.00 (3.96%)	12.00 (5.80%)	
	Black	29.00 (9.42%)	10.00 (9.90%)	19.00 (9.18%)	
	Hispanic	8.00 (2.60%)	2.00 (1.98%)	6.00 (2.90%)	
	Other	29.00 (9.42%)	7.00 (6.93%)	22.00 (10.63%)	
	White	226.00 (73.38%)	78.00 (77.23%)	148.00 (71.50%)	
Weight		72.50 (60.98, 85.90)	69.10 (63.00, 85.00)	74.33 (60.15, 86.45)	0.810
HR		94.00 (82.50, 109.50)	98.00 (86.00, 113.00)	91.00 (80.00, 106.00)	0.006
NBPS		112.00 (100.00, 126.00)	109.00 (101.00, 123.00)	112.00 (99.00, 129.00)	0.454
NBPD		64.00 (54.00, 73.00)	66.00 (53.00, 74.00)	63.00 (54.00, 71.00)	0.493
RR		20.00 (16.00, 23.50)	20.00 (17.00, 25.00)	20.00 (16.00, 23.00)	0.158
Temperature		36.78 (36.50, 37.06)	36.72 (36.44, 37.06)	36.78 (36.50, 37.06)	0.451
WBC		12.00 (7.55, 17.55)	13.70 (8.70, 20.20)	10.90 (7.20, 16.10)	0.008
PlateletCount		180.50 (114.00, 254.00)	172.00 (98.00, 244.00)	186.00 (121.00, 263.00)	0.164
Hemoglobin		9.40 (8.40, 11.00)	9.30 (8.40, 11.30)	9.40 (8.40, 10.90)	0.677
Sodium		137.00 (133.50, 139.00)	136.00 (132.00, 139.00)	137.00 (134.00, 139.00)	0.093
Potassium		4.00 (3.60, 4.50)	4.10 (3.70, 5.00)	3.90 (3.60, 4.30)	0.004
CalciumTotal		8.00 (7.40, 8.50)	7.90 (7.40, 8.60)	8.00 (7.40, 8.50)	0.883
Chloride		103.00 (99.00, 107.00)	101.00 (97.00, 107.00)	104.00 (100.00, 107.00)	0.014
BilirubinTotal		1.20 (0.60, 3.80)	1.40 (0.80, 4.80)	1.10 (0.60, 2.90)	0.011
ALT		45.00 (20.00, 97.00)	45.00 (22.00, 102.00)	44.00 (20.00, 97.00)	0.544
AST		61.00 (29.00, 119.00)	76.00 (38.00, 150.00)	52.00 (26.00, 106.00)	0.007
UreaNitrogen		20.50 (13.00, 35.50)	30.00 (17.00, 54.00)	17.00 (12.00, 29.00)	<0.001
Creatinine		0.90 (0.70, 1.55)	1.10 (0.70, 1.90)	0.90 (0.70, 1.40)	0.007
Hypertension					0.976
	No	162.00 (52.60%)	53.00 (52.48%)	109.00 (52.66%)	
	Yes	146.00 (47.40%)	48.00 (47.52%)	98.00 (47.34%)	
T2DM					0.773
	No	158.00 (51.30%)	53.00 (52.48%)	105.00 (50.72%)	
	Yes	150.00 (48.70%)	48.00 (47.52%)	102.00 (49.28%)	
T1DM					>0.999
	No	307.00 (99.68%)	101.00 (100.00%)	206.00 (99.52%)	
	Yes	1.00 (0.32%)	0.00 (0.00%)	1.00 (0.48%)	
Heart Failure					0.122
	No	261.00 (84.74%)	81.00 (80.20%)	180.00 (86.96%)	
	Yes	47.00 (15.26%)	20.00 (19.80%)	27.00 (13.04%)	
Myocardial Infarction					0.162
	No	289.00 (93.83%)	92.00 (91.09%)	197.00 (95.17%)	
	Yes	19.00 (6.17%)	9.00 (8.91%)	10.00 (4.83%)	
CKD					0.528
	No	256.00 (83.12%)	82.00 (81.19%)	174.00 (84.06%)	
	Yes	52.00 (16.88%)	19.00 (18.81%)	33.00 (15.94%)	
Pneumonia					0.107
	No	231.00 (75.00%)	70.00 (69.31%)	161.00 (77.78%)	
	Yes	77.00 (25.00%)	31.00 (30.69%)	46.00 (22.22%)	
Stroke					0.399
	No	294.00 (95.45%)	95.00 (94.06%)	199.00 (96.14%)	
	Yes	14.00 (4.55%)	6.00 (5.94%)	8.00 (3.86%)	
Hyperlipidemia					0.945
	No	196.00 (63.64%)	64.00 (63.37%)	132.00 (63.77%)	
	Yes	112.00 (36.36%)	37.00 (36.63%)	75.00 (36.23%)	
COPD					0.156
	No	293.00 (95.13%)	99.00 (98.02%)	194.00 (93.72%)	
	Yes	15.00 (4.87%)	2.00 (1.98%)	13.00 (6.28%)	
RAR		6.37 (5.22, 7.80)	6.89 (5.77, 8.32)	6.17 (5.13, 7.26)	<0.001

Continuous variables are presented as median (interquartile range).

Categorical variables are presented as count (percentage).

Statistical comparisons were performed using the Wilcoxon rank-sum test or Pearson's chi-squared test.

Table 1-eICU.Baseline Characteristics					
Variable		Overall N = 141	Non-survivor N = 19	Survivor N = 122	p-value
Age		66.20 (10.40)	67.47 (9.22)	66.00 (10.60)	0.631
Gender					0.936
	Female	73.00 (51.77%)	10.00 (52.63%)	63.00 (51.64%)	0.532
	Male	68.00 (48.23%)	9.00 (47.37%)	59.00 (48.36%)	
Race					0.532
	Asian	3.00 (2.13%)	0.00 (0.00%)	3.00 (2.46%)	
	Black	17.00 (12.06%)	3.00 (15.79%)	14.00 (11.48%)	
	Hispanic	11.00 (7.80%)	3.00 (15.79%)	8.00 (6.56%)	
	Other	3.00 (2.13%)	0.00 (0.00%)	3.00 (2.46%)	
	White	107.00 (75.89%)	13.00 (68.42%)	94.00 (77.05%)	
Weight		74.90 (61.60, 89.00)	79.00 (61.60, 89.00)	74.40 (61.00, 90.00)	0.726
HR		89.00 (80.00, 104.00)	103.00 (85.00, 113.00)	89.00 (80.00, 102.00)	0.045
NBPS		112.00 (98.00, 134.00)	93.00 (88.00, 114.00)	116.50 (101.00, 135.00)	0.017
NBPD		64.50 (17.00)	66.47 (21.10)	64.20 (16.36)	0.983
RR		18.84 (5.04)	21.58 (6.08)	18.42 (4.75)	0.031
Temperature		36.70 (36.40, 37.10)	36.70 (36.40, 36.90)	36.70 (36.40, 37.10)	0.555
WBC		12.20 (8.20, 16.90)	16.80 (6.70, 26.80)	11.85 (8.20, 15.40)	0.043
PlateletCount		182.00 (123.00, 239.00)	152.00 (75.00, 239.00)	184.00 (126.00, 243.00)	0.272
Hemoglobin		9.93 (1.87)	9.29 (1.69)	10.03 (1.89)	0.107
Sodium		138.00 (135.00, 141.00)	135.00 (133.00, 143.00)	138.00 (136.00, 141.00)	0.350
Potassium		4.10 (3.70, 4.50)	4.40 (3.70, 5.10)	4.10 (3.70, 4.40)	0.056
CalciumTotal		7.86 (0.68)	7.76 (0.84)	7.88 (0.65)	0.103
Chloride		105.00 (103.00, 109.00)	105.00 (99.00, 109.00)	106.00 (103.00, 110.00)	0.336
BilirubinTotal		1.20 (0.60, 3.10)	2.10 (1.00, 3.30)	1.10 (0.60, 3.10)	0.074
ALT		53.00 (28.00, 110.00)	59.00 (32.00, 530.00)	52.00 (27.00, 108.00)	0.146
AST		61.00 (29.00, 125.00)	85.00 (39.00, 1,369.00)	60.50 (28.00, 118.00)	0.054
UreaNitrogen		16.00 (11.00, 29.00)	30.00 (17.00, 56.00)	15.00 (11.00, 27.00)	<0.001
Creatinine		0.80 (0.60, 1.40)	1.72 (0.94, 2.43)	0.78 (0.59, 1.14)	<0.001
Hypertension					0.530
	No	117.00 (82.98%)	17.00 (89.47%)	100.00 (81.97%)	>0.999
	Yes	24.00 (17.02%)	2.00 (10.53%)	22.00 (18.03%)	
T2DM					>0.999
	No	137.00 (97.16%)	19.00 (100.00%)	118.00 (96.72%)	
	Yes	4.00 (2.84%)	0.00 (0.00%)	4.00 (3.28%)	>0.999
T1DM					
	No	139.00 (98.58%)	19.00 (100.00%)	120.00 (98.36%)	0.186
	Yes	2.00 (1.42%)	0.00 (0.00%)	2.00 (1.64%)	
Heart Failure					>0.999
	No	135.00 (95.74%)	17.00 (89.47%)	118.00 (96.72%)	
	Yes	6.00 (4.26%)	2.00 (10.53%)	4.00 (3.28%)	0.348
Myocardial Infarction					
	No	139.00 (98.58%)	19.00 (100.00%)	120.00 (98.36%)	0.348
	Yes	2.00 (1.42%)	0.00 (0.00%)	2.00 (1.64%)	
CKD					>0.999
	No	132.00 (93.62%)	17.00 (89.47%)	115.00 (94.26%)	
	Yes	9.00 (6.38%)	2.00 (10.53%)	7.00 (5.74%)	>0.999
Pneumonia					
	No	131.00 (92.91%)	18.00 (94.74%)	113.00 (92.62%)	>0.999
	Yes	10.00 (7.09%)	1.00 (5.26%)	9.00 (7.38%)	
Stroke					0.594
	No	140.00 (99.29%)	19.00 (100.00%)	121.00 (99.18%)	
	Yes	1.00 (0.71%)	0.00 (0.00%)	1.00 (0.82%)	0.075
Hyperlipidemia					
	No	134.00 (95.04%)	19.00 (100.00%)	115.00 (94.26%)	<0.001
	Yes	7.00 (4.96%)	0.00 (0.00%)	7.00 (5.74%)	
COPD					<0.001
	No	133.00 (94.33%)	16.00 (84.21%)	117.00 (95.90%)	
	Yes	8.00 (5.67%)	3.00 (15.79%)	5.00 (4.10%)	
RAR		6.68 (5.56, 8.53)	9.04 (7.29, 10.32)	6.38 (5.23, 8.04)	<0.001
Continuous variables are presented as median (interquartile range).					
Categorical variables are presented as count (percentage).					
Statistical comparisons were performed using the Wilcoxon rank-sum test or Pearson's chi-squared test.					

解读：死亡组患者表现出更明显的高代谢应激与脏器损伤迹象——心率更快、白细胞更高、血钾升高、氯偏低、胆红素与 AST 升高、尿素氮明显升高。合并症负担在两组间相近，提示在本队列中**急性生理紊乱**而非**慢性基础疾病**驱动了短期死亡。RAR 在死亡组显著升高，与上述趋势一致，支持其作为风险标志物的合理性。

3.3 单因素分析：Table S2

Table S2. Analysis of Univariable（MIMIC、eICU 分库）。

Table S1. Analysis of Univariable				
name	desc	Survivor (N=207)	Non-survivor (N=101)	OR (univariable)
Age	Median (IQR)	69.0 (62.0 to 77.0)	72.0 (64.0 to 78.0)	1.01 (0.99-1.03, p=0.241)
Gender	Female	95 (45.9%)	38 (37.6%)	1.41 (0.86-2.29, p=0.170)
	Male	112 (54.1%)	63 (62.4%)	
Race	Asian	12 (5.8%)	4 (4%)	1.58 (0.40-6.19, p=0.512)
	Black	19 (9.2%)	10 (9.9%)	
	Hispanic	6 (2.9%)	2 (2%)	
	Other	22 (10.6%)	7 (6.9%)	
	White	148 (71.5%)	78 (77.2%)	
Weight	Median (IQR)	74.3 (60.5 to 86.3)	69.1 (63.0 to 85.0)	1.00 (0.99-1.01, p=0.912)
HR	Median (IQR)	91.0 (80.5 to 106.0)	98.0 (86.0 to 113.0)	1.02 (1.00-1.03, p=0.012)
NBPS	Median (IQR)	112.0 (99.5 to 129.0)	109.0 (101.0 to 123.0)	0.99 (0.98-1.00, p=0.273)
NBPD	Median (IQR)	63.0 (54.0 to 70.5)	66.0 (53.0 to 74.0)	1.00 (0.99-1.02, p=0.867)
RR	Median (IQR)	20.0 (16.0 to 22.5)	20.0 (17.0 to 25.0)	1.04 (0.99-1.08, p=0.086)
Temperature	Median (IQR)	36.8 (36.5 to 37.1)	36.7 (36.4 to 37.1)	0.88 (0.59-1.31, p=0.518)
WBC	Median (IQR)	10.9 (7.2 to 16.0)	13.7 (8.7 to 20.2)	1.03 (1.00-1.05, p=0.033)
PlateletCount	Median (IQR)	186.0 (121.0 to 260.0)	172.0 (98.0 to 244.0)	1.00 (1.00-1.00, p=0.220)
Hemoglobin	Median (IQR)	9.4 (8.4 to 10.9)	9.3 (8.4 to 11.3)	1.05 (0.93-1.19, p=0.459)
Sodium	Median (IQR)	137.0 (134.0 to 139.0)	136.0 (132.0 to 139.0)	0.96 (0.92-1.00, p=0.071)
Potassium	Median (IQR)	3.9 (3.6 to 4.3)	4.1 (3.7 to 5.0)	1.87 (1.35-2.60, p<0.001)
CalciumTotal	Median (IQR)	8.0 (7.4 to 8.5)	7.9 (7.4 to 8.6)	1.12 (0.85-1.48, p=0.416)
Chloride	Median (IQR)	104.0 (100.0 to 107.0)	101.0 (97.0 to 107.0)	0.96 (0.92-0.99, p=0.018)
BilirubinTotal	Median (IQR)	1.1 (0.6 to 2.8)	1.4 (0.8 to 4.8)	1.05 (1.00-1.10, p=0.044)
ALT	Median (IQR)	44.0 (20.0 to 95.5)	45.0 (22.0 to 102.0)	1.00 (1.00-1.00, p=0.342)
AST	Median (IQR)	52.0 (26.5 to 105.0)	76.0 (38.0 to 150.0)	1.00 (1.00-1.00, p=0.024)
UreaNitrogen	Median (IQR)	17.0 (12.0 to 29.0)	30.0 (17.0 to 54.0)	1.02 (1.01-1.03, p<0.001)
Creatinine	Median (IQR)	0.9 (0.7 to 1.4)	1.1 (0.7 to 1.9)	1.13 (0.96-1.34, p=0.136)
Hypertension	No	109 (52.7%)	53 (52.5%)	1.01 (0.63-1.62, p=0.976)
	Yes	98 (47.3%)	48 (47.5%)	
T2DM	No	105 (50.7%)	53 (52.5%)	0.93 (0.58-1.50, p=0.773)
	Yes	102 (49.3%)	48 (47.5%)	
T1DM	No	206 (99.5%)	101 (100%)	0.00 (0.00-Inf, p=0.988)
	Yes	1 (0.5%)	0 (0%)	
Heart Failure	No	180 (87%)	81 (80.2%)	1.65 (0.87-3.11, p=0.124)
	Yes	27 (13%)	20 (19.8%)	
Myocardial Infarction	No	197 (95.2%)	92 (91.1%)	1.93 (0.76-4.90, p=0.169)
	Yes	10 (4.8%)	9 (8.9%)	
CKD	No	174 (84.1%)	82 (81.2%)	1.22 (0.66-2.28, p=0.528)
	Yes	33 (15.9%)	19 (18.8%)	
Pneumonia	No	161 (77.8%)	70 (69.3%)	1.55 (0.91-2.65, p=0.108)
	Yes	46 (22.2%)	31 (30.7%)	
Stroke	No	199 (96.1%)	95 (94.1%)	1.57 (0.53-4.66, p=0.415)
	Yes	8 (3.9%)	6 (5.9%)	
Hyperlipidemia	No	132 (63.8%)	64 (63.4%)	1.02 (0.62-1.67, p=0.945)
	Yes	75 (36.2%)	37 (36.6%)	
COPD	No	194 (93.7%)	99 (98%)	0.30 (0.07-1.36, p=0.119)
	Yes	13 (6.3%)	2 (2%)	
RAR	Median (IQR)	6.2 (5.1 to 7.2)	6.9 (5.8 to 8.3)	1.18 (1.06-1.32, p=0.003)

Table S1. Analysis of Univariable				
name	desc	Survivor (N=122)	Non-survivor (N=19)	OR (univariable)
Age	Median (IQR)	66.5 (58.0 to 74.0)	65.0 (61.0 to 72.5)	1.01 (0.97-1.06, p=0.565)
Gender	Female	63 (51.6%)	10 (52.6%)	
	Male	59 (48.4%)	9 (47.4%)	0.96 (0.37-2.53, p=0.936)
Race	Asian	3 (2.5%)	0 (0%)	
	Black	14 (11.5%)	3 (15.8%)	9116746.91 (0.00-Inf, p=0.994)
	Hispanic	8 (6.6%)	3 (15.8%)	15954307.09 (0.00-Inf, p=0.994)
	Other	3 (2.5%)	0 (0%)	1.00 (0.00-Inf, p=1.000)
	White	94 (77%)	13 (68.4%)	5883857.93 (0.00-Inf, p=0.995)
Weight	Median (IQR)	74.4 (61.4 to 89.7)	79.0 (62.2 to 89.0)	1.01 (0.99-1.03, p=0.538)
HR	Median (IQR)	89.0 (80.0 to 102.0)	103.0 (85.5 to 111.0)	1.03 (1.00-1.06, p=0.042)
NBPS	Median (IQR)	116.5 (101.0 to 135.0)	93.0 (88.5 to 113.0)	0.97 (0.95-1.00, p=0.018)
NBPD	Median (IQR)	62.5 (53.2 to 76.0)	61.0 (52.5 to 71.0)	1.01 (0.98-1.04, p=0.586)
RR	Median (IQR)	18.0 (15.0 to 22.0)	20.0 (18.0 to 26.5)	1.13 (1.03-1.25, p=0.013)
Temperature	Median (IQR)	36.7 (36.4 to 37.1)	36.7 (36.5 to 36.8)	0.55 (0.30-1.02, p=0.057)
WBC	Median (IQR)	11.9 (8.3 to 15.4)	16.8 (9.5 to 24.5)	1.08 (1.02-1.14, p=0.009)
PlateletCount	Median (IQR)	184.0 (127.0 to 241.8)	152.0 (91.5 to 235.5)	1.00 (0.99-1.00, p=0.187)
Hemoglobin	Median (IQR)	9.9 (8.6 to 11.5)	9.5 (8.2 to 10.8)	0.80 (0.61-1.05, p=0.113)
Sodium	Median (IQR)	138.0 (136.0 to 141.0)	135.0 (133.0 to 143.0)	0.96 (0.87-1.06, p=0.431)
Potassium	Median (IQR)	4.1 (3.7 to 4.4)	4.4 (3.9 to 5.0)	1.95 (0.99-3.85, p=0.053)
CalciumTotal	Median (IQR)	7.9 (7.5 to 8.3)	7.7 (7.3 to 7.8)	0.77 (0.37-1.57, p=0.464)
Chloride	Median (IQR)	106.0 (103.0 to 109.8)	105.0 (100.5 to 108.5)	0.95 (0.88-1.02, p=0.168)
BilirubinTotal	Median (IQR)	1.1 (0.6 to 3.1)	2.1 (1.1 to 3.1)	1.02 (0.90-1.14, p=0.770)
ALT	Median (IQR)	52.0 (27.2 to 107.5)	59.0 (37.0 to 356.0)	1.00 (1.00-1.00, p=0.066)
AST	Median (IQR)	60.5 (28.2 to 116.8)	85.0 (43.0 to 1061.5)	1.00 (1.00-1.00, p=0.005)
UreaNitrogen	Median (IQR)	15.0 (11.0 to 27.0)	30.0 (18.0 to 54.5)	1.03 (1.01-1.05, p=0.002)
Creatinine	Median (IQR)	0.8 (0.6 to 1.1)	1.7 (1.1 to 2.4)	1.56 (1.16-2.10, p=0.003)
Hypertension	No	100 (82%)	17 (89.5%)	
	Yes	22 (18%)	2 (10.5%)	0.53 (0.12-2.48, p=0.424)
T2DM	No	118 (96.7%)	19 (100%)	
	Yes	4 (3.3%)	0 (0%)	0.00 (0.00-Inf, p=0.990)
T1DM	No	120 (98.4%)	19 (100%)	
	Yes	2 (1.6%)	0 (0%)	0.00 (0.00-Inf, p=0.993)
Heart Failure	No	118 (96.7%)	17 (89.5%)	
	Yes	4 (3.3%)	2 (10.5%)	3.47 (0.59-20.41, p=0.169)
Myocardial Infarction	No	120 (98.4%)	19 (100%)	
	Yes	2 (1.6%)	0 (0%)	0.00 (0.00-Inf, p=0.993)
CKD	No	115 (94.3%)	17 (89.5%)	
	Yes	7 (5.7%)	2 (10.5%)	1.93 (0.37-10.08, p=0.434)
Pneumonia	No	113 (92.6%)	18 (94.7%)	
	Yes	9 (7.4%)	1 (5.3%)	0.70 (0.08-5.84, p=0.740)
Stroke	No	121 (99.2%)	19 (100%)	
	Yes	1 (0.8%)	0 (0%)	0.00 (0.00-Inf, p=0.993)
Hyperlipidemia	No	115 (94.3%)	19 (100%)	
	Yes	7 (5.7%)	0 (0%)	0.00 (0.00-Inf, p=0.992)
COPD	No	117 (95.9%)	16 (84.2%)	
	Yes	5 (4.1%)	3 (15.8%)	4.39 (0.96-20.14, p=0.057)
RAR	Median (IQR)	6.4 (5.2 to 8.0)	9.0 (7.5 to 10.0)	1.70 (1.29-2.23, p<0.001)

训练集（MIMIC）中与死亡显著相关的变量（ $p < 0.05$ ）：HR、WBC、钾、氯、总胆红素、AST、尿素氮、RAR。年龄、性别及各并发症均未达显著。

RAR 的单因素结果：

- **MIMIC**：存活 6.2 (5.1–7.2) vs 死亡 6.9 (5.8–8.3)，OR 1.18 (95% CI 1.06–1.32)， $p = 0.003$ 。

- **eICU**：存活 6.4 (5.2–8.0) vs 死亡 9.0 (7.5–10.0)，OR 1.70 (95% CI 1.29–2.23)， $p < 0.001$ 。

解读：训练集筛选出 8 个与结局相关的连续变量，加之年龄、性别作为常规人口学校正变量，构成后续建模的核心特征集。值得注意的是，RAR 在两库中方向一致、效应稳健，且在样本更小、死亡率更低的验证集中效应更强（OR 1.70），提示 RAR 的预后价值并非训练集特有。验证集中肌酐、RR、收缩压亦显著，但为避免信息泄露，这些变量的纳入与否由训练集结果决定。

3.4 共线性诊断：Table S3 与 Fig 2

Table S3-MIMIC. Variance Inflation Factor (VIF) Analysis		Table S3-eICU. Variance Inflation Factor (VIF) Analysis	
Variable	VIF	Variable	VIF
Age	1.169	Age	1.39
Gender	1.044	Gender	1.895
HR	1.393	HR	1.244
WBC	1.26	WBC	1.871
Potassium	1.52	Potassium	1.334
Chloride	1.075	Chloride	1.41
BilirubinTotal	1.74	BilirubinTotal	1.87
AST	1.126	AST	1.573
UreaNitrogen	1.139	UreaNitrogen	1.607
RAR	1.089	RAR	1.258

Table S3. Variance Inflation Factor (VIF) Analysis。

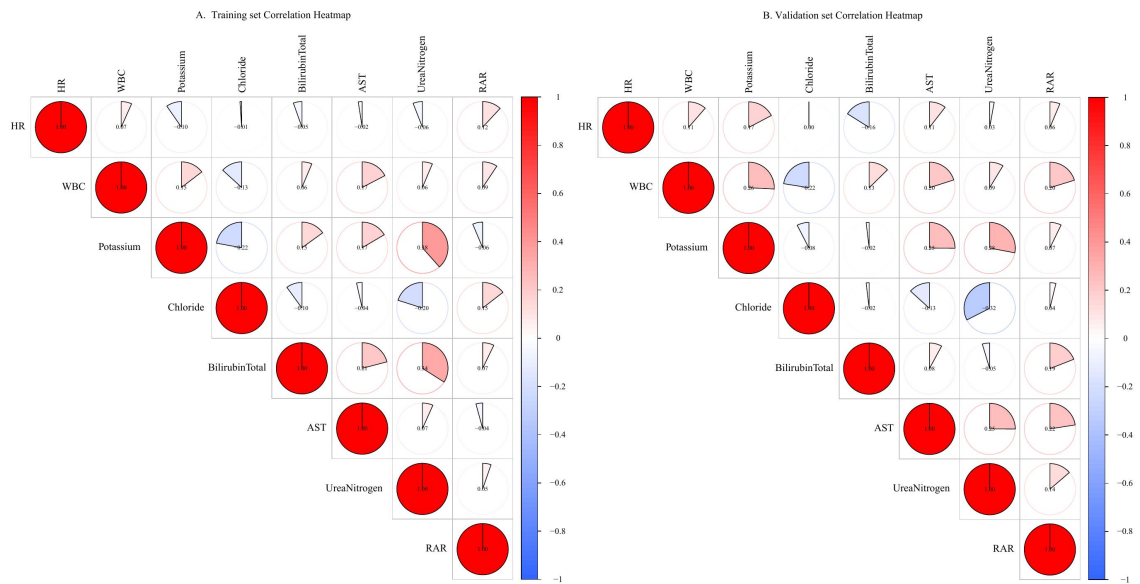


Fig 2. Correlation heatmap of the candidate continuous features in the training (A) and validation (B) sets.

结果：所有入选变量的 VIF 均 < 2（最大为总胆红素 1.74），远低于通常的阈值 5 或 10。相关性热图（Fig 2）显示，训练集与验证集中两两变量间相关系数普遍较低，未出现强共线性区块。

解读：纳入模型的特征彼此独立，不存在严重的多重共线性。这意味着后续逻辑回归与 Cox 回归的系数估计不会被变量间相关所扭曲，RAR 的独立效应可以被可靠地分离出来。

3.5 RAR 与结局的关联：Fig 3、Table 2、Table 3、Fig 4

4.5.1 剂量-反应关系：Fig 3

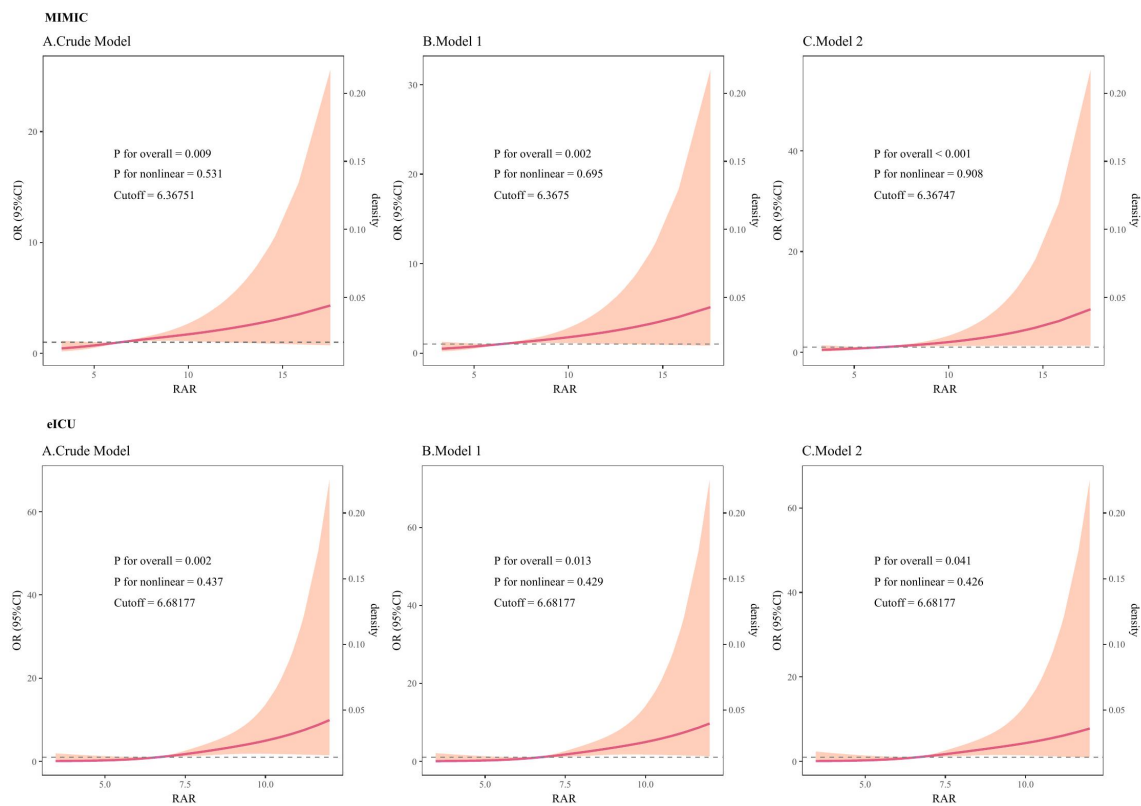


Fig 3. Restricted cubic spline (RCS) showing the association between RAR and 28-day in-hospital mortality in MIMIC (A–C) and eICU (A–C), under the crude model, Model 1 and Model 2. OR with 95% CI; reference set at the lowest risk.

Model 1 调整年龄、性别、HR；*Model 2* 进一步调整WBC、钾、氯、总胆红素、AST、尿素氮。

解读：在两库的全部模型中，RAR 与死亡的整体关联均显著（ P for overall ≤ 0.041 ），而非线性检验均不显著（ $P > 0.4$ ），说明 RAR 与死亡呈**近似线性的剂量-反应关系**——风险随 RAR 升高而持续增加，并未出现明显的“断点效应”或 U 型拐点。RCS 推导的最优切点在 MIMIC 为 6.37、eICU 为 6.68，与两库 RAR 的中位数（6.37 与 6.68）高度吻合，提示切点稳定、具生物学合理性。

4.5.2 逻辑回归（OR）：Table 2

Table 2. Logistic regression analysis of RAR（MIMIC、eICU 分库）。

Table2-MIMIC. Logistic regression analysis of RAR										
Characterisitic	Exposure cutoff	Crude Model			Model1			Model2		
		OR	95%CI	P-value	OR	95%CI	P-value	OR	95%CI	P-value
RAR										
RAR continuous		1.184	(1.062,1.325)	0.0027	1.185	(1.06,1.332)	0.0033	1.22	(1.082,1.383)	0.0014
RAR category										
Lower RAR	< 6.36	Ref	Ref		Ref	Ref		Ref	Ref	
Higher RAR	>= 6.36	1.869	(1.156,3.048)	0.0113	1.795	(1.094,2.97)	0.0214	2.027	(1.184,3.515)	0.0107
Crude Model was non-adjusted										
The Model 1 was adjusted by Age,Gender,HR										
The Model 2 was adjusted by Age,Gender,HR,WBC,Potassium,Chloride,BilirubinTotal,AST,UreaNitrogen										

Table2-eICU. Logistic regression analysis of RAR										
Characterisitic	Exposure cutoff	Crude Model			Model1			Model2		
		OR	95%CI	P-value	OR	95%CI	P-value	OR	95%CI	P-value
RAR										
RAR continuous		1.697	(1.31,2.276)	1e-04	1.701	(1.3,2.309)	2e-04	1.648	(1.24,2.278)	0.0011
RAR category										
Lower RAR	< 6.68	Ref	Ref		Ref	Ref		Ref	Ref	
Higher RAR	>= 6.68	4.42	(1.504,16.194)	0.012	4.425	(1.476,16.475)	0.0133	3.813	(1.193,14.763)	0.0327
Crude Model was non-adjusted										
The Model 1 was adjusted by Age,Gender,HR										
The Model 2 was adjusted by Age,Gender,HR,WBC,Potassium,Chloride,BilirubinTotal,AST,UreaNitrogen										

切点：MIMIC 6.36，eICU 6.68。

解读：RAR 每升高一个单位，死亡风险的调整后 OR 在 MIMIC 约 1.22、在 eICU 约 1.65；按切点二分类后，高 RAR 组死亡风险在 MIMIC 约为低组的 2 倍、在 eICU 约为 3.8 倍。逐级调整混杂后效应基本不变甚至略有增强，提示 RAR 的关联不易被已知混杂解释。验证集效应更强，可能与该队列高 RAR 患者比例更高、分布更偏右有关。

4.5.3 Cox 回归（HR）与生存曲线：Table 3、Fig 4

Table 3-MIMIC. The Association Between RAR										
Characterisitic	Exposure cutoff	Crude Model			Model1			Model2		
		HR	95%CI	P-value	HR	95%CI	P-value	HR	95%CI	P-value
RAR										
RAR continuous		1.673	(1.122,2.493)	0.0115	1.575	(1.05,2.364)	0.0281	1.631	(1.077,2.469)	0.0208
RAR category										
Lower RAR	< 6.36	Ref			Ref			Ref		
Higher RAR	>= 6.36	1.439	(1.085,1.908)	0.0115	1.379	(1.035,1.837)	0.0281	1.413	(1.054,1.895)	0.0208
Crude Model was non-adjusted										
The Model 1 was adjusted by Age,Gender,HR										
The Model 2 was adjusted by Age,Gender,HR,WBC,Potassium,Chloride,BilirubinTotal,AST,UreaNitrogen										

Table 3-eICU. The Association Between RAR										
Characterisitic	Exposure cutoff	Crude Model			Model1			Model2		
		HR	95%CI	P-value	HR	95%CI	P-value	HR	95%CI	P-value
RAR										
RAR continuous		3.553	(1.176,10.737)	0.0247	3.585	(1.182,10.874)	0.0241	3.631	(1.019,12.939)	0.0467
RAR category										
Lower RAR	< 6.68	Ref			Ref			Ref		
Higher RAR	>= 6.68	2.451	(1.121,5.357)	0.0247	2.467	(1.126,5.405)	0.0241	2.489	(1.013,6.113)	0.0467
Crude Model was non-adjusted										
The Model 1 was adjusted by Age,Gender,HR										
The Model 2 was adjusted by Age,Gender,HR,WBC,Potassium,Chloride,BilirubinTotal,AST,UreaNitrogen										

Table 3. The association between RAR and 28-day mortality（Cox 回归，HR）。

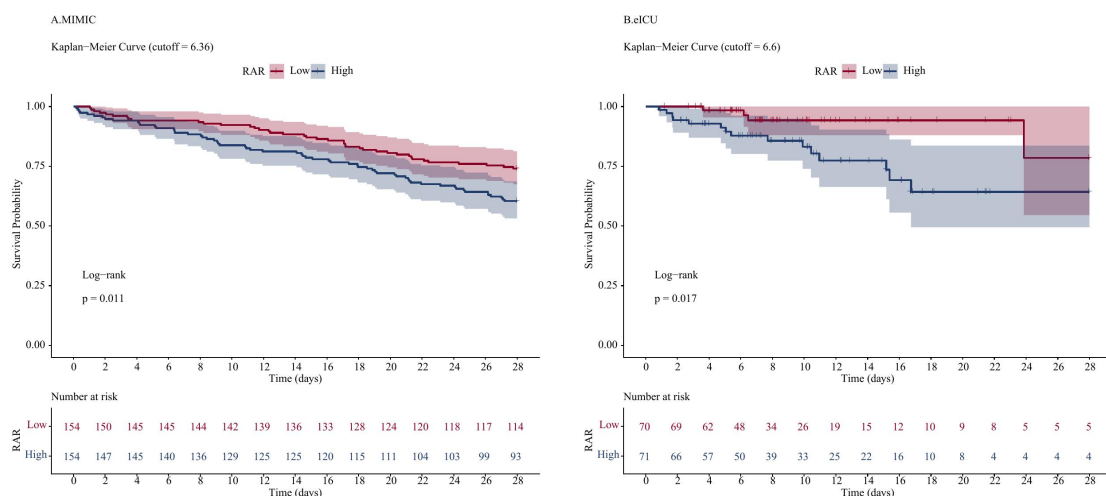


Fig 4. Kaplan-Meier survival curves stratified by RAR group (high vs low) in MIMIC (A, cutoff = 6.36) and eICU (B, cutoff = 6.6); log-rank test.

结果：Cox 回归与逻辑回归结论一致，RAR 升高伴随更高的瞬时死亡风险，调整后仍显著。Kaplan-Meier 曲线显示，高 RAR 组在两库中的生存率均自早期即低于低 RAR 组，曲线持续分离、无交叉；log-rank 检验 MIMIC $p = 0.011$ ，eICU $p = 0.017$ 。

解读：从累积生存（KM）与瞬时风险（Cox）两个角度，RAR 的预后价值都得到印证。曲线早期分离提示 RAR 反映的是入院时已存在的、与短期死亡直接相关的生理负荷，而非迟发事件。结合 RCS 的线性结论，可认为 RAR 对死亡风险的影响贯穿整个 28 天观察窗口。

3.6 LASSO 变量筛选：Fig 5

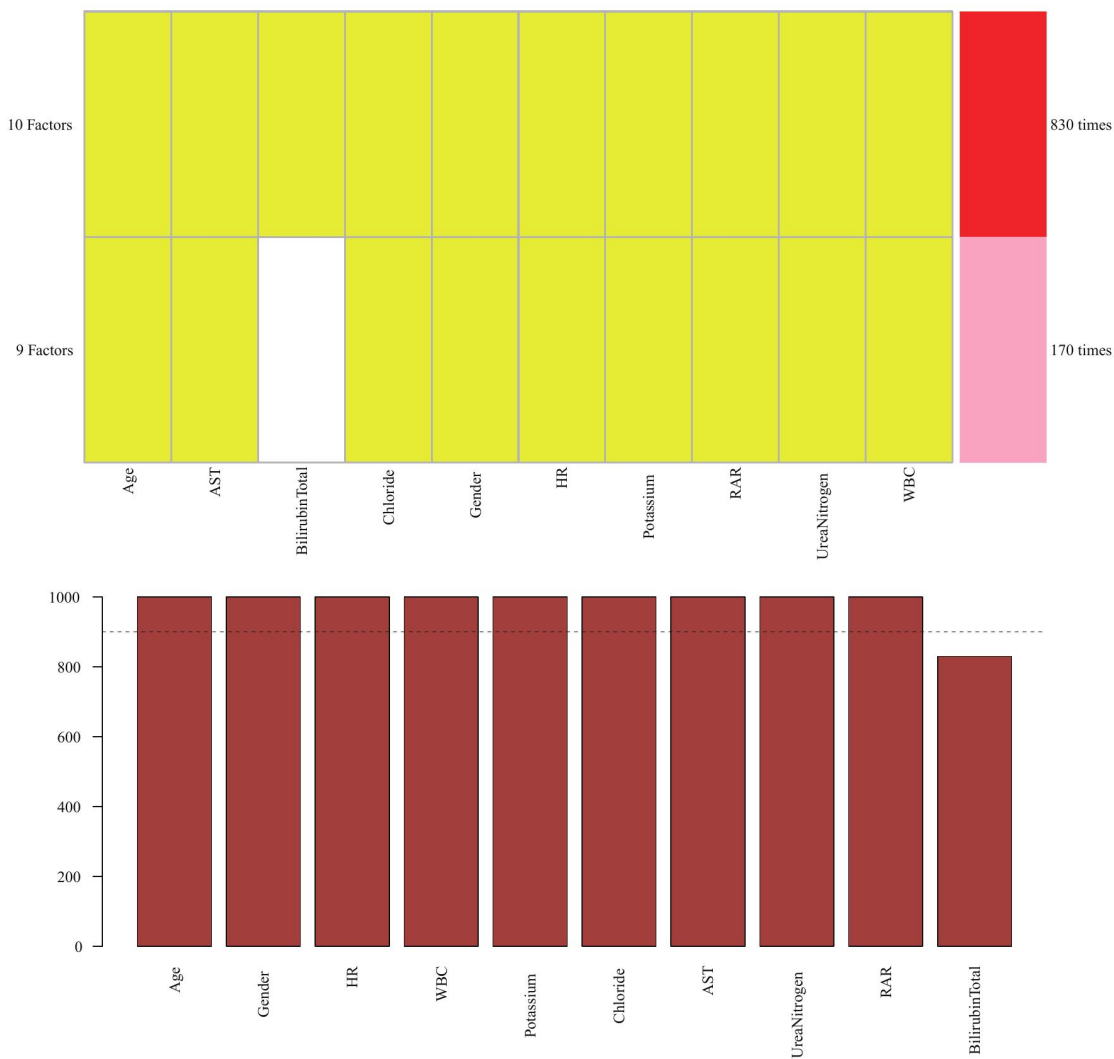


Fig 5. LASSO-based stability selection: inclusion frequency of each candidate feature across bootstrap resamples in the training set.

结果：在训练集上采用基于 LASSO 的自助法稳定性选择（bootstrap stability selection）。候选特征为年龄、性别、HR、WBC、钾、氯、总胆红素、AST、尿素氮、RAR 共 10 个。在重复抽样中，10 变量解出现约 830 次、9 变量解约 170 次；各变量被选中的频率均处于高位，无变量被系统性剔除。

解读：稳定性选择的结果支持保留全部 10 个特征进入机器学习模型。这一结果与单因素筛选、共线性诊断相互印证——这些变量既与结局相关、彼此独立，又在重抽样中表现稳定，适合作为最终的建模特征集。

3.7 训练集与验证集基线：Table S4

Table S4. Baseline Characteristics （训练集 vs 验证集）。

Table S4.Baseline Characteristics				
Variable		Overall N = 449	Train N = 308	Validation N = 141
DN	Survivor	329.00 (73.27%)	207.00 (67.21%)	122.00 (86.52%)
	Non-survivor	120.00 (26.73%)	101.00 (32.79%)	19.00 (13.48%)
Age		68.59 (11.12)	69.68 (11.28)	66.20 (10.40)
Gender	Female	206.00 (45.88%)	133.00 (43.18%)	73.00 (51.77%)
	Male	243.00 (54.12%)	175.00 (56.82%)	68.00 (48.23%)
HR		93.00 (82.00, 106.00)	94.00 (82.50, 109.50)	89.00 (80.00, 104.00)
WBC		12.00 (7.70, 17.10)	12.00 (7.55, 17.55)	12.20 (8.20, 16.90)
Potassium		4.00 (3.60, 4.50)	4.00 (3.60, 4.50)	4.10 (3.70, 4.50)
Chloride		104.00 (100.00, 108.00)	103.00 (99.00, 107.00)	105.00 (103.00, 109.00)
BilirubinTotal		1.20 (0.60, 3.30)	1.20 (0.60, 3.80)	1.20 (0.60, 3.10)
AST		61.00 (29.00, 121.00)	61.00 (29.00, 119.00)	61.00 (29.00, 125.00)
UreaNitrogen		19.00 (12.00, 33.00)	20.50 (13.00, 35.50)	16.00 (11.00, 29.00)
RAR		6.42 (5.27, 7.94)	6.37 (5.22, 7.80)	6.68 (5.56, 8.53)
Continuous variables are presented as median (interquartile range).				
Categorical variables are presented as count (percentage).				

解读：两库存在可预期的差异。验证集（eICU）死亡率更低（13.5% vs 32.8%）、年龄略小、RAR 中位数略高。这种基线异质性使外部验证更具挑战性，也更具说服力：若模型在此条件下仍保持判别力，说明其泛化能力较为稳健。同时，类别不平衡的差异（训练集正例约 1:2，验证集约 1:6）也解释了为何部分模型在验证集中的敏感度下降。

3.8 机器学习模型构建与评价：Fig 6、Table 4/5、Table S5–S10

共训练 15 种模型，涵盖梯度提升类（XGBoost、LightGBM、CatBoost）、集成树（RF）、核方法（RSVM）、神经网络（MLP、RealMLP）、表格基础模型（TabPFN、TabPFNv2、RealTabPFN-2.5、TabICL v2）、近邻（KNN）、逻辑回归与弹性网（ENet）、决策树（DT）。超参数见表 S5。

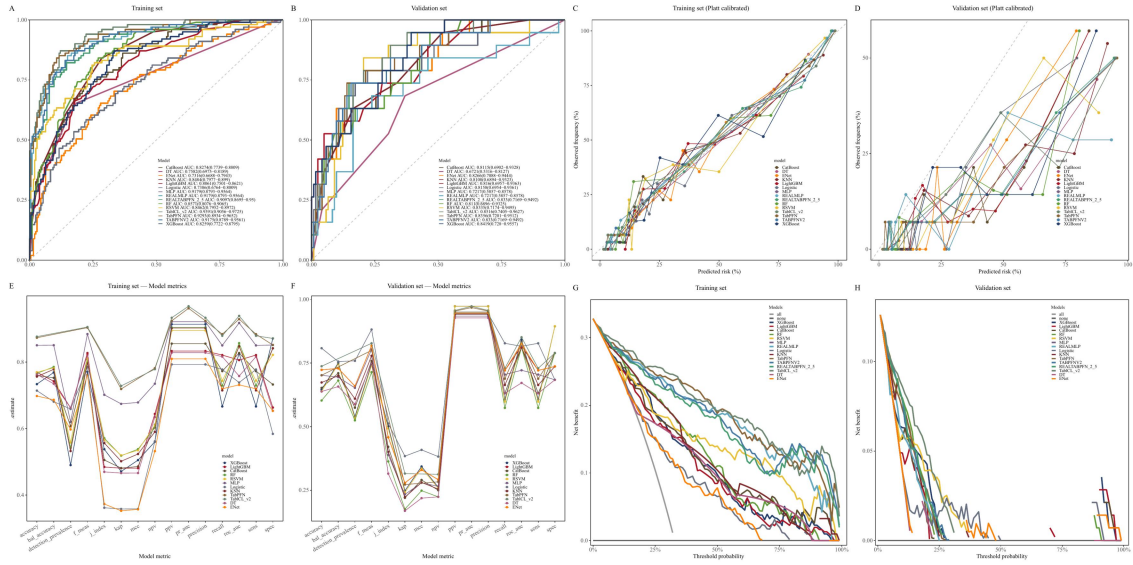


Fig 6. Machine-learning model performance. (A,B) ROC curves in training and validation; (C,D) calibration curves (Platt); (E,F) metric bar charts; (G,H) decision-curve analysis.

Table 4/5（节选，按验证集 AUC 排序）

Table 4. ML model performance (training) — Pancreatic Cancer					
Model	AUC	accuracy	Sensitivity	Specificity	F1
XGBoost	0.826	0.734	0.667	0.871	0.771
LightGBM	0.806	0.769	0.821	0.663	0.827
CatBoost	0.827	0.76	0.773	0.733	0.812
RF	0.857	0.766	0.729	0.842	0.807
RSVM	0.846	0.769	0.744	0.822	0.813
MLP	0.918	0.851	0.85	0.851	0.884
RealMLP	0.918	0.851	0.85	0.851	0.884
Logistic	0.739	0.714	0.778	0.584	0.785
KNN	0.848	0.756	0.715	0.842	0.798
TabPFN	0.929	0.873	0.884	0.851	0.904
TabPFNv2	0.917	0.838	0.821	0.871	0.872
RealTabPFN-2.5	0.91	0.844	0.855	0.822	0.881
TabCL v2	0.939	0.877	0.879	0.871	0.905
DT	0.758	0.763	0.816	0.653	0.822
ENet	0.732	0.698	0.72	0.653	0.762

Table 5. ML model performance (validation) — Pancreatic Cancer					
Model	AUC	accuracy	Sensitivity	Specificity	F1
XGBoost	0.842	0.645	0.607	0.895	0.747
LightGBM	0.816	0.723	0.721	0.737	0.819
CatBoost	0.811	0.652	0.631	0.789	0.759
RF	0.811	0.603	0.574	0.789	0.714
RSVM	0.833	0.638	0.598	0.895	0.741
MLP	0.722	0.702	0.705	0.684	0.804
RealMLP	0.722	0.702	0.705	0.684	0.804
Logistic	0.816	0.809	0.828	0.684	0.882
KNN	0.811	0.674	0.664	0.737	0.779
TabPFN	0.836	0.702	0.689	0.789	0.8
TabPFNv2	0.833	0.681	0.656	0.842	0.78
RealTabPFN-2.5	0.833	0.709	0.705	0.737	0.808
TabCL v2	0.852	0.738	0.73	0.789	0.828
DT	0.672	0.638	0.631	0.684	0.751
ENet	0.827	0.723	0.721	0.737	0.819

两两比较与增量价值 (Table S7–S10)： DeLong 检验显示，在训练集中表格基础模型与 MLP 显著优于 XGBoost、逻辑回归等（如 TablCL v2 vs XGBoost, $p < 0.001$ ）；但在验证集中这一差距明显收窄，XGBoost、RSVM、TabPFN、TablCL v2 的 AUC 落在同一区间（0.834–0.852），两两差异多不再显著。NRI/IDI 分析中，强模型相对 ENet、DT 等弱基线可获得显著的整合判别改善（如 TablCL v2 vs ENet, IDI 0.128, $p < 0.001$ ），而强模型之间的净重分类多无显著差异。校准曲线（Platt 标定）显示主要模型预测概率与实测发生率贴合良好；决策曲线分析提示在常用风险阈值区间，主要模型相对"全部干预"或"均不干预"均存在正的净获益。

解读： 第一，验证集中表现最佳的是 TablCL v2（AUC 0.852），但梯度提升（XGBoost 0.842）与核方法（RSVM 0.834）紧随其后，提示在中等样本量下，结构相对稳健的模型泛化更可靠。第二，高容量模型（MLP、TabPFN 系列部分变体）训练集 AUC 接近 0.92，验证集却回落至 0.72–0.84，存在明显过拟合，说明此类模型在小样本表格数据上需谨慎使用。第三，逻辑回归与弹性网在验证集中 AUC 同样达到 0.82 左右，且校准与 F1 表现不差，提示特征集本身已具备较强的判别力，模型的边际贡献有限。第四，校准与决策曲线表明模型具备潜在的临床实用价值，但需结合本地患病率重新标定阈值。

3.9 模型可解释性：Fig 7

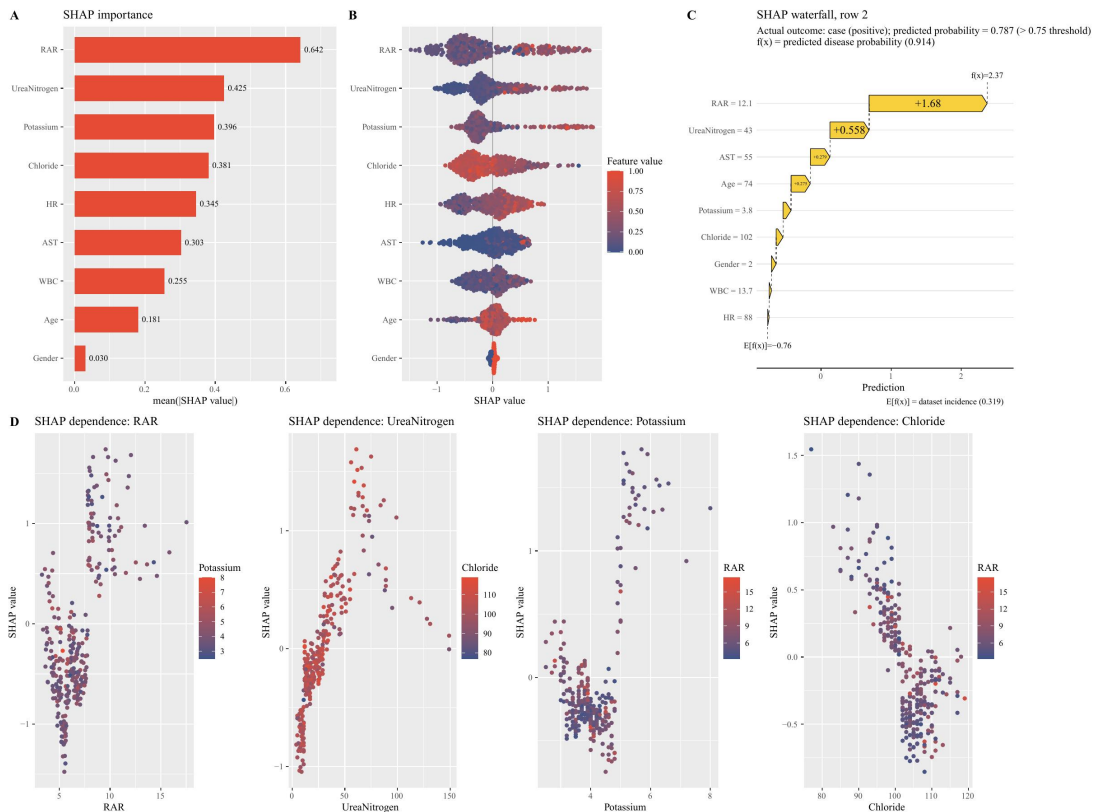


Fig 7. SHAP-based global and local interpretation of the best-performing model. (A) Global feature importance; (B) SHAP summary (beeswarm); (C) local waterfall for an individual patient; (D) dependence plots for RAR, UreaNitrogen, Potassium, Chloride.

结果：全局重要性排序（平均 | SHAP | ）依次为：RAR（0.642）> 尿素氮（0.425）> 钾（0.396）> 氯（0.381）> HR（0.345）> AST（0.303）> WBC（0.255）> 年龄（0.181）> 性别（0.030）。蜂群图显示，RAR、尿素氮、钾、氯、HR 取值越高（红色），SHAP 值越偏向正，即对死亡预测的正向贡献越大。依赖图中 RAR 的 SHAP 值随其取值升高而持续上升，无明显平台或转折。局部瀑布图给出了一例高风险患者的解释：基线期望 $E[f(x)] = 0.319$ ，最终预测概率 0.787，其中 RAR 贡献 +1.68、尿素氮 +0.558，是将其判为高危的主要因素。

解读：SHAP 分析把“黑箱”模型的决策落到可读的变量层面。RAR 排名第一，远超第二位的尿素氮，说明模型对死亡的判别在很大程度上由 RAR 主导，这与前述单因素、逻辑回归、Cox 的结论完全一致——多种分析框架独立指向同一变量，增强了结论的稳健性。依赖图进一步验证了 RCS 的线性结论：RAR 越高、风险越高，二者形态吻合。局部解释则展示了模型如何在个体层面给出可追溯的判断，为临床沟通与质控提供了依据。

四、主要发现小结

- 1. RAR 与 28 天院内死亡独立相关：**在训练集与验证集中均显著，调整多重混杂后效应不减；剂量-反应近似线性，非线性检验不显著。
- 2. 切点稳定：**RCS 推导切点（MIMIC 6.37、eICU 6.68）与各自中位数高度一致，便于临床转化。
- 3. 特征集精简且无共线性：**10 个特征通过单因素、VIF、稳定性选择三重检验，构成稳健的建模基础。
- 4. 模型判别力可接受：**验证集中 TablCL v2、XGBoost、RSVM、TabPFN 的 AUC 约 0.83–0.85；高容量模型存在过拟合。
- 5. RAR 是首要驱动变量：**SHAP 全局排名第一，与统计学分析结论一致，模型可解释性强。